

A Brief Experimental Survey of Semantic Image Segmentation on PanNuke and NuCLS datasets

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Abstract—Semantic image segmentation is changing digital pathology by making it faster and more accurate to analyze tissue samples through digital images. Whole Slide Imaging converts these samples into high-resolution digital formats, allowing for detailed analysis. Traditional semantic segmentation uses deep learning models, but evaluations have mainly focused on specific datasets, limiting broader applicability. This paper presents a brief experimental survey of semantic image segmentation on two well-known datasets: PanNuke and NuCLS. We assess the performance of traditional deep learning models on these datasets, highlighting recent trends and challenges. Our experimental survey aims to help the understanding of semantic image segmentation and eventually to the field of digital pathology advance, making diagnosis more accurate and accessible. Our findings can guide researchers and clinicians in selecting appropriate models for their specific needs, ultimately improving patient care through more precise disease detection.

Index Terms—Semantic Image Segmentation, Deep Learning, Histopathology Image Analysis, Instance-Based Nuclei Segmentation, Convolutional Neural Networks

I. INTRODUCTION

Semantic image segmentation is an ever-evolving field of study that has become increasingly popular in recent years because of its ability to segment objects in complex images quickly and more efficiently. Deep learning inspired by some functionalities of the human brain's neural networks have been applied to the semantic image segmentation to provide a significant improvement by taking sufficient training data and recognizing patterns to predict task intent. The ability to perform human-like tasks has enabled semantic image segmentation equipped with deep learning-based approaches to be adopted in various studies such as computer vision, medical sciences, and robotics.

Whole slide imaging is a field of imaging technology that produces high-resolution digital images from the microscopic slide, fosters in pathologists a rapid progress towards work at the generation of a large volume of image data that is highly reliable and useful in determining treatment by clinical doctors. Whole-slide imaging technologies have enabled microscope equipment to digitize an entire glass slide within a few minutes, while early microscope scanners have provided over 24 hours of scanning a single slide, transforming the field of digital pathology during the last few years. Now digital

pathology data have become increasingly common in clinical and basic science investigations.

Deep learning models, such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs), have been successfully employed in the detection, diagnosis, treatment, and prevention of human cancers, aiming to promote the transition of digital pathology [5]. These models for analyzing tissue samples obtained for microscopic examination provide satisfactory outcomes and validated clinical prediction while specialized software tools with onboard deep learning-based methods assure faster turnaround times in a quantitative analysis of whole slide images.

In this paper, we present a brief experimental survey evaluating state-of-the-art deep learning models for instance segmentation including semantic image segmentation and cell nuclei detection on hematoxylin and eosin-stained histopathology images. The first contribution of the paper is providing a review of state-of-the-art convolutional neural network architectures tailored for instance segmentation, including MaskRCNN, Detectron2, HoverNet, YOLOv8, and YOLOv9. Next, we examine the significance of preprocessing techniques for histopathology images, such as color normalization methods to mitigate staining variability and data augmentation strategies to increase dataset diversity. Last, we evaluate the performance of state-of-the-art instance segmentation models on two popular datasets: the PanNuke and NuCLS, which contain annotated nuclei from a wide range of cancer types. We perform the experiments on both binary and multiclass instance segmentation for both datasets. The PanNuke dataset offers annotations across 19 tissue types, while the NuCLS dataset provides detailed labeling of nuclei in breast cancer histopathology images.

II. METHOD

We initially perform the preprocessing of H&E stained images as the quality of these techniques significantly impacts result accuracy. Recent advancements have introduced various preprocessing methods, including normalization, augmentation, registration, and segmentation.

Normalization techniques aim to reduce the variability in image intensity and enhance the contrast [7].

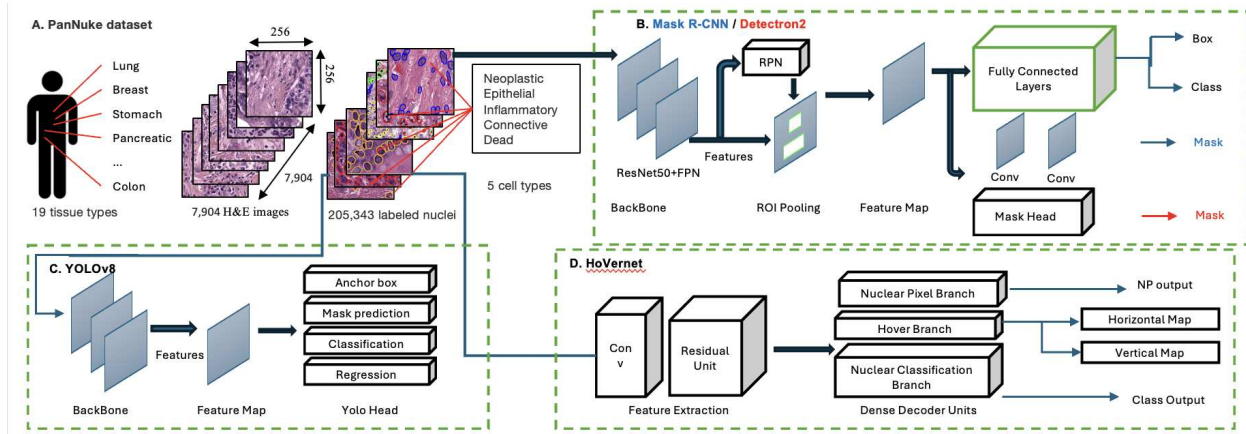


Fig. 1. Overview of the PanNuke Dataset and models used in this study. (A) The PanNuke dataset comprises 7,904 H&E-stained images from 19 tissue types with 205,343 labeled nuclei across five cell classes: neoplastic, epithelial, inflammatory, connective, and dead. (B) Mask R-CNN with Detectron2 framework is used for multi-class segmentation with a ResNet50+FPN backbone and ROI pooling for feature extraction, followed by fully connected layers for box, class, and mask predictions. (C) YOLOv8 model uses an anchor box and YOLO head for single-class segmentation tasks. (D) HoverNet includes a feature extraction block with nuclear pixel, hover, and classification branches to generate nuclear and class outputs.

Segmentation techniques separate the tissue structures from the background and identify the regions of interest [6]. One of the key preprocessing techniques for H&E stained images is color normalization, which separates individual stain components (hematoxylin and eosin) from the RGB image using techniques like color deconvolution [1]. In this paper, we utilized two datasets, NuCLS and PanNuke, to explore both binary and multiclass instance segmentation approaches. For the NuCLS dataset, models such as Mask R-CNN, Detectron2, YOLOv8, and YOLOv9 were employed for multiclass instance segmentation across multiple cell types. The PanNuke dataset was used for both multiclass and single binary instance segmentation. Multiclass instance segmentation was achieved with HoverNet and Mask R-CNN, while single binary instance segmentation utilized Mask R-CNN, Detectron2, YOLOv8, and YOLOv9. Each ground truth mask contained only a single cell nucleus of one type among five distinct classes across 19 tissue types. The details of our experiments are shown in Fig. 1.

MaskRCNN with a ResNet-101 backbone is a powerful instance segmentation model that combines deep convolutional networks for robust feature extraction with a Region Proposal Network and RoI Align for precise object localization. This architecture enables delineation and classification of objects, making it suitable for complex computer vision tasks [8].

Detectron2, built on the PyTorch framework, implements advanced models like Mask R-CNN with a ResNet50 backbone and Feature Pyramid Network (FPN). This combination allows for efficient multi-scale feature extraction and object detection across various sizes and resolutions [9].

YOLOv8, the iteration in the YOLO series, is a single-stage detector that predicts bounding boxes and segmentation masks in one pass through the network. It utilizes an EfficientRep backbone along with specialized YOLO heads for fast and

efficient object detection and instance segmentation [12].

YOLOv9 introduces innovations such as Programmable Gradient Information (PGI) and Reversible Functions to address information bottlenecks in deep neural networks. These advancements, coupled with an efficient backbone architecture like DRNets, enhance information flow and gradient generation, resulting in a state-of-the-art real-time performance for object detection and segmentation [10].

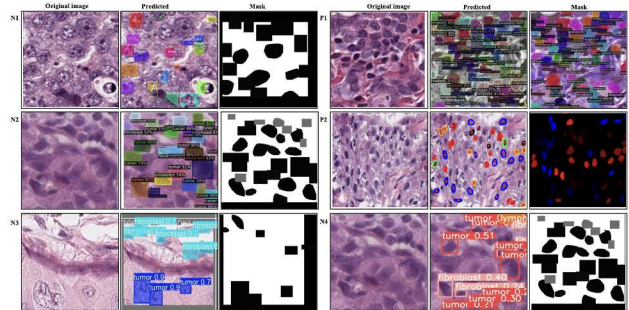


Fig. 2. Comparison of Instance Segmentation Models on the NuCLS and PanNuke Datasets. Figures N1, N2, N3, and N4 show the instance segmentation performance of four models—Mask R-CNN (N1), Detectron2 (N2), YOLOv8 (N3), and YOLOv9 (N4)—on the NuCLS dataset. Each row includes the original histopathology image, the model's predicted segmentation with cell-type labels, and the corresponding binary mask, highlighting how each model identifies and segments various cell types. Figures P1 and P2 compare Mask R-CNN and HoverNet on the PanNuke dataset, with similar layouts. MaskRCNN uses bounding boxes to label various cell types, while HoverNet outlines individual cell nuclei with color-coded contours for different types. The figure highlights each model's capability to segment and identify cell nuclei in complex histopathology images.

HoverNet is a specialized deep-learning model designed for nuclear segmentation in histopathological images. Its dual-branch architecture simultaneously segments and classifies nuclei, producing both segmentation masks and nuclear classi-

TABLE I
PERFORMANCE METRICS FOR VARIOUS MODELS ON NUCLS AND PANNUKE DATASETS

Model	Precision	Recall	F1 Score	Accuracy	IoU	Dice Coefficient	mAP
MaskRCNN (NuCLS)	0.7857	0.5303	0.6332	0.7000	0.6500	0.7500	0.3000
Detectron2 (NuCLS)	0.5404	0.6816	0.5956	0.6818	0.6000	0.7000	0.3156
YOLOv8 (NuCLS)	0.8180	0.4920	0.6150	0.7500	0.7000	0.8000	0.3580
YOLOv9 (NuCLS)	0.5720	0.2860	0.3800	0.6500	0.5500	0.6500	0.4570
MaskRCNN (PanNuke)	0.7500	0.6000	0.6700	0.8000	0.7000	0.7800	0.1880
HoverNet (PanNuke)	0.8000	0.7000	0.7500	0.9421	0.7500	0.8310	0.4000

fication information. This unique approach allows HoverNet to excel in dense prediction tasks and handle multi-class segmentation effectively, particularly in complex histopathological images [2].

III. EXPERIMENTAL RESULTS

Dataset: Two well-known datasets were selected for our experiment. The NuCLS (Nuclei Classification) dataset is a comprehensive collection of breast cancer histopathology images, featuring over 220,000 manually annotated nuclei instances [11]. It features high-resolution images of 13 types and is organized into training, validation, and test subsets. Its diversity reflects the complexities of real-world histopathology, providing single-rater and multi-rater data with annotations in various formats for inter-rater analysis and annotation improvement methods[3, 4]. The PanNuke dataset is designed for advanced nuclei instance segmentation and classification research across various cancerous tissues. It includes 205,343 labeled nuclei instances from 481 visual fields derived from over 20,000 whole-slide images, covering 19 distinct tissue types. Each type is categorized into five specific nuclei categories: neoplastic, inflammatory, soft tissue, dead, and epithelial, allowing for detailed analysis of nuclei types within different tissue contexts.

Evaluation metrics: To evaluate deep-learning models, we used the Intersection over Union (IoU), Dice Coefficient, Precision, Recall, F-score, Average Precision (AP), Mean Average Precision (mAP), and Accuracy. We denote the AP and mAP as follows:

AP assesses the detection quality of each class. It is computed as the area under the Precision-Recall curve:

$$AP = \int_0^1 \text{Precision}(r) dr \quad (1)$$

where r represents recall. AP provides an overall score for each class in the model.

mAP measures the average AP across all classes in multi-class tasks, providing a single metric for overall performance:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (2)$$

where N is the total number of classes.

While intuitive, accuracy can be misleading in imbalanced datasets, such as those with a dominant background class. The combined use of these metrics provides a thorough

understanding of our models' strengths and limitations in nuclei classification tasks, essential for practical applications in histopathological analysis.

Results: Our study evaluated the performance of various deep learning models for semantic image segmentation on the PanNuke and NuCLS datasets. For PanNuke, we conducted both multi-class and binary instance segmentation experiments. In multi-class segmentation, HoverNet outperformed Mask R-CNN across all metrics. For binary segmentation, YOLOv8 emerged as the top performer among Mask R-CNN, Detectron2, YOLOv8, and YOLOv9, excelling particularly in segmenting neoplastic cells across various tissue types as shown in Table II and Fig. 3.

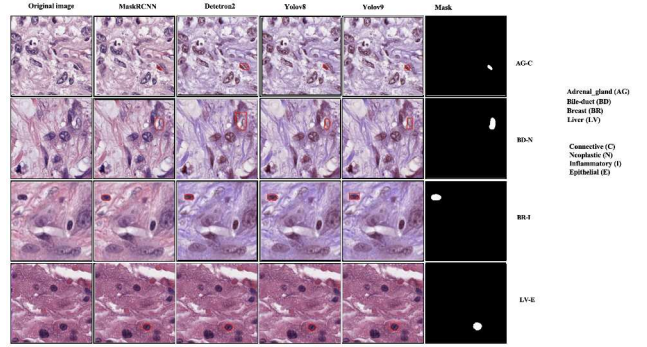


Fig. 3. Single instance segmentation of the PanNuke dataset was performed using Mask R-CNN, YOLOv8, YOLOv9, and Detectron2. Each row showcases examples from different tissue types and cell classes, including Adrenal Gland-Connective (AG-C), Bile Duct-Neoplastic (BD-N), Breast-Inflammatory (BR-I), and Liver-Epithelial (LV-E). The first column displays the original histopathology image, followed by predictions from each model and the corresponding binary mask.

On the NuCLS dataset, which focused on multi-class instance segmentation, YOLOv8 again outperformed the other models, with Mask R-CNN and YOLOv9 showing comparable results. Mask R-CNN performed well in certain scenarios, especially in segmenting inflammatory cells in some tissue types, while Detectron2 excelled in specific tissue types and cell categories, particularly segmenting neoplastic cells in tissues like skin and bile duct. The comparison of instance segmentation models on two datasets is shown in Table I and Fig. 2. Performance varied across different tissue types and cell categories, highlighting the importance of model selection based on specific histopathology tasks.

TABLE II
COMPARISON OF INSTANCE SEGMENTATION MODELS CLASSIFICATION

Adrenal gland														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9091	0.9304	0.9534	0.8813	0.7339	0.7664	0.8975	0.73	0.8581	0.8547	0.9173	0.8101	0.9544	0.808
mAP	0.8735	0.8591	0.9622	0.8942	0.8648	0.7537	0.8863	0.799	0.7840	0.8766	0.8783	0.8307	0.9614	0.808
Bile-duct														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9735	0.9657	0.9502	0.8853	0.9303	0.9005	0.9338	0.8634	0.9065	0.9213	0.9454	0.7949	0.8589	0.94
mAP	0.9047	0.9413	0.9774	0.9114	0.8370	0.9463	0.9403	0.9011	0.8357	0.581	0.9675	0.8	0.8654	0.935
Breast														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9523	0.9576	0.9535	0.9011	0.9647	0.9263	0.931	0.8502	0.9469	0.9512	0.9285	0.8468	0.9540	0.9527
mAP	0.9773	0.885	0.9832	0.9254	0.8254	0.8361	0.949	0.908	0.8102	0.8792	0.9334	0.902	0.8441	0.8205
Cervix														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9371	0.9516	0.9529	0.8832	0.9356	0.9232	0.9305	0.8506	0.9067	0.9094	0.9032	0.8588	0.9562	0.94
mAP	0.9064	0.9036	0.938	0.8662	0.9248	0.9271	0.949	0.8553	0.9181	0.8619	0.9131	0.8415	0.9581	0.91
Esophagus														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9354	0.9052	0.95	0.9277	0.8443	0.7954	0.9151	0.7925	0.9037	0.9245	0.8474	0.8560	0.5660	0.496
mAP	0.8049	0.9061	0.9015	0.9132	0.8679	0.8334	0.9101	0.8858	0.8122	0.9142	0.9404	0.8596	0.5660	0.496
Kidney														
Neoplastic					Inflammatory				Connective				Epithelial	
	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	Detecron2	YOLOv8	YOLOv9	MaskRCNN	YOLOv8
AP	0.9587	0.9134	0.8361	0.8515	0.3050	0.65	0.70	0.68	0.7473	0.7832	0.6238	0.711	0.6245	0.80
mAP	0.8261	0.9474	0.8583	0.8825	0.0000	0.60	0.65	0.63	0.7473	0.7832	0.7042	0.7663	0.6245	0.75

IV. CONCLUSION

In conclusion, this paper surveys recent advancements in histopathology image analysis using deep learning algorithms. Our experimental study on the PanNuke and NuCLS datasets revealed varying model performance across different tissue types and cell categories. On the PanNuke dataset, YOLOv8 consistently outperformed other models, particularly in segmenting neoplastic cells across various tissue types. However, Mask R-CNN and Detecron2 demonstrated strengths in specific tasks, such as inflammatory cell segmentation. In the NuCLS dataset, YOLOv8 again showed strong performance with high precision and Dice coefficients, while HoverNet excelled in recall and overall accuracy. These findings underscore the importance of model selection based on specific histopathology tasks and tissue types. Our work aims to guide researchers and clinicians in selecting appropriate models for enhanced diagnostic accuracy and patient care in digital pathology.

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